

Math 526, S16

Quiz 2

Section:

Name: *Answer*

SSN: xxx-xx-

Instructions: There are totally 2 problems. You must show all of your work to receive a credit for a correct answer.

Problem 1. (4 points) Compute the multiplication of the two matrices

$$\begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -1 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 0 \\ 1 & 3 & 1 \end{bmatrix}.$$
$$= \begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & -6 \\ 0 & 1 & -2 \end{bmatrix}$$

Problem 2. (6 points) Write down the 3×3 matrices that produce the following operations:

(a) E_{21} subtracts $1/2$ times row 1 from row 2.

$$E_{21} = \begin{bmatrix} 1 & 0 & 0 \\ -\frac{1}{2} & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

(b) E_{32} adds 7 times row 2 to row 3.

$$E_{32} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 7 & 1 \end{bmatrix}$$

(c) P_{13} exchanges row 1 and row 3.

$$P_{13} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$$